IN THE STEIN FACTORIZATION OF A UNIVERSALLY CLOSED FINITE-TYPE MAP OF SPACES TO AN AFFINE BASE, THE FIBRES OF PI ARE FINITE WITH FINITE RESIDUE EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8060-8066 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8068-8088, the complete author proof (19 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The author explicitly writes 'A topological argument (omitted)' for the finiteness of the point set; everything else is written out with exact citations.

Editorial note. Counted B side; A (schemes) twin uncounted per ruling.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. In-source lemma-stein-universally-closed; Algebra, lemma integral-no-inclusion; Morphisms of Spaces, lemmas finite-presentation-noetherian and enough-finite-type-points; More on Morphisms, lemma point-finite-type.

Source reference. Original lemma 00GT: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8088–8088. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03GY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7994–8014. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01TA: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8088–8088. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06EK: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8088–8088. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04ZL: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8088–8088. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Lemma 03GY assume in addition that f is locally of finite type and Y affine. Then for $y \in Y$ the fibre $\pi^{-1}(\{y\}) = \{y_1, \ldots, y_n\}$ is finite and the field extensions $\kappa(y_i)/\kappa(y)$ are finite.*

Proof. Recall that there are no specializations among the points of $\pi^{-1}(\{y\})$, see Algebra, Lemma 00GT. As f' is surjective, we find that $|X_y| \to \pi^{-1}(\{y\})$ is surjective. Observe that X_y is a quasi-separated algebraic space of finite type over a field (quasi-compactness was shown in the proof of the referenced lemma). Thus $|X_y|$ is a Noetherian topological space (Morphisms of Spaces, Lemma 04ZL). A topological argument (omitted) now shows that $\pi^{-1}(\{y\})$ is finite. For each i we can pick a finite type point $x_i \in |X_y|$ mapping to y_i (Morphisms of Spaces, Lemma 06EK). We conclude that $\kappa(y_i)/\kappa(y)$ is finite: x_i can be represented by a morphism $\mathrm{Spec}(k_i) \to X_y$ of finite type (by our definition of finite type points) and hence $\mathrm{Spec}(k_i) \to y = \mathrm{Spec}(\kappa(y))$ is of finite type (as a composition of finite type morphisms), hence $k_i/\kappa(y)$ is finite (Morphisms, Lemma 01TA). □